

Physiology lec 6

Autonomic Nervous system (ANS)

- Before we talk about ANS we should know that nervous system is divided into **Central nervous system (CNS)** and **Peripheral nervous system (PNS)**
- ANS and somatic nervous system are derived from PNS
- While in CNS consist of Brain and spinal cord
- spinal cord contain **cervical, thoracic, lumbar and sacral nerves**

Nervous system (NS) actions

- 1-Sensory by sense organs such as ears and eyes
 - 2-Motor action which involves **somatic NS** and **ANS**
- Somatic NS is the one that we control like skeletal muscles
 - ANS is the involuntary one (uncontrolled) like heart beats and blood vessels

ANS Controls the following

1-Smooth muscles in viscerals (احشاء) such as the Peristalsis (which is similar to worm movement)

2-Cardiac muscles to control heart beats

3-glands and their secretions

4-reflex actions

---ANS consist of 2 systems Sympathetic nervous system and Parasympathetic nervous system

--Those two systems their function antagonize the other (عكس بعض)

Before The systems lets talk about ganglia

--When Cell bodies of Nerve cells gather together making a collection this is called ganglia

--formed outside NS and not formed in brain / spinal cord

---Presence: ANS in both sympathetic and parasympathetic

---consist of 2 parts

1-preganglionic which synapse with Nerves and give it the impulse

2-Postganglionic which takes impulse from nerve and give it to organ

Preganglionic → Nerve → postganglionic → Organ

N.B: a- Preganglionic (Myelinated) is **shorter than postganglionic (Unmyelinated) in length in Sympathetic NS**

Function of Gangolia???

1-Collect multiple branches of nerves

2-absorb shocks

3-Weakens nerve signal if it is too strong

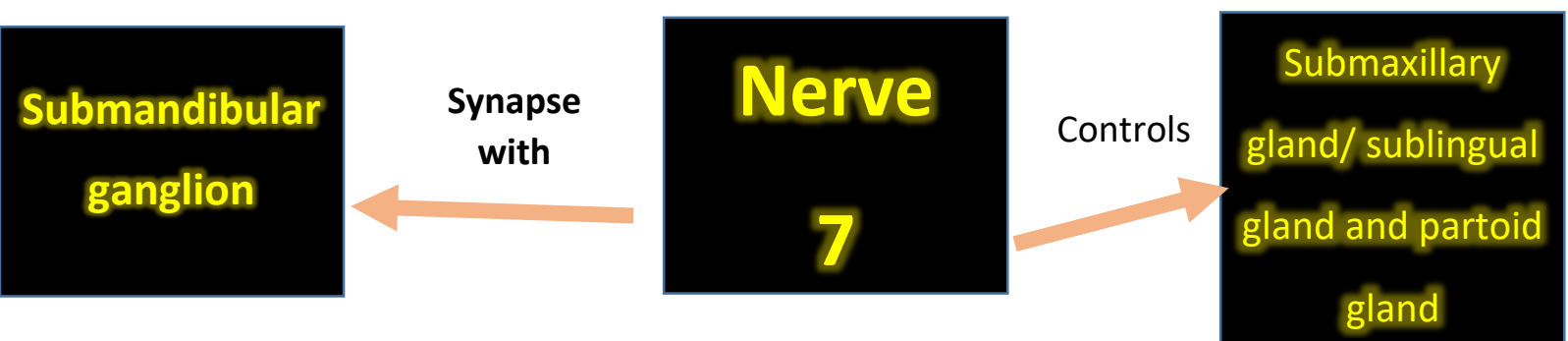
And strengthen it if too weak (Optimization**)**

Firstly: Parasympathetic

---Another term for parasympathetic is craniosacral system,,, because its cells are located in brain stem and sacral region of spinal cord

---Parasympathetic secretes acetylcholine for its actions

And The parasympathetic nerves are 3/7/9/10



**Pterygopalatine
ganglion**

Synapse
with

Nerve 7

Controls

Lacrimal Gland / Mucous membrane (Mucous is in nose and palate)

Lacrimal Gland is responsible for crying

**Vagus
Nerve**

Synapse
with

**Nerve
10**

Controls

**Heart / Larynx / Trachea / Bronchi (in lungs) / esophagus / stomach / abdomen
blood vessels / liver / pancreas / adrenal gland / small and large intestine and
rectum / kidney / ureter / urinary bladder / gonads / external genitalia**

Otic Ganglion

Synapse
with

**Nerve
9**

Controls

**Mucous
membrane which
is in mouth**

Sympathetic Nervous System

---Known as thoracolumbar system

----it goes to same places parasympathetic goes to such as eyes/ stomach / kidney and etc...

But nerves coming from spinal cord goes into 2 directions

While nerves coming from brain goes into 1 direction

Sympathetic Nervous system is responsible for secretion of Adrenaline (Epinephrine) & Nor adrenaline (Nor Epinephrine)

Autonomic Reflex??

It is a reflex when organ send a signal to NS (here NS called afferent because it receives signal) then signal is translated and send in form of impulses to organ (here organ is called efferent because it will cause motion or action)

Types of Reflexes

1-Cardiovascular Autonomic reflexes

Controls blood pressure / heart rate and cardiac output

Parasympathetic Decrease heart rate

Sympathetic Increase heart rate

-Blood pressure is changed through Baroreceptors

Where when pressure **decrease** baroreceptors send signals to **Sympathetic NS** to increase

When pressure **increase** baroreceptors send signal to **Parasympathetic NS** to decrease

2nd type of Autonomic Reflexes is

Gastrointestinal autonomic reflex

-Signal between stomach & intestine & salivary gland which controls secretions and motility (movement)

--Motility increase by Parasympathetic

----Motility Decrease by Sympathetic

Last Type of Autonomic Reflexes is sexual reflexes

--Controls Sexual Activity

**--It receives stimuli from brain and sex organ
IN male**

Parasympathetic is responsible for erection

Sympathetic is responsible for ejaculation

Made By: Malak Ezzat

Gehad Metwally

Karim Maged